

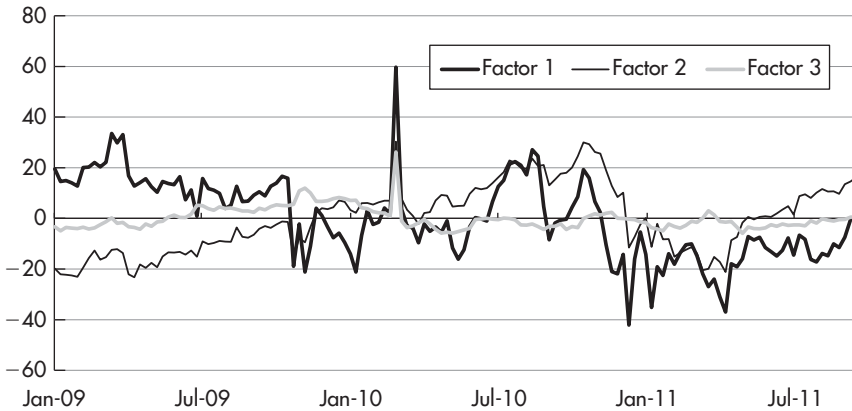


FIGURE 19.25 First three factors of a PCA on the vega sector of the JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

relative value trades on factor 2. (See Chapter 3.) Thus in Japan, factor 3 is the only source for good relative value trades on the volatility surface, exhibiting a high speed of mean reversion and little correlation to factor 1 or 2 during subperiods. A trade on factor 3 is a position on the volatility *curve* (same expiry, different underlying swap maturities), which requires *three* instruments, a fact that may seem a bit puzzling at first. Again, the situation is different in other markets, where both factors 2 and 3 could offer relative value trades on the volatility surface with a high speed of mean reversion.

As always, one can link the PCA factors to external explaining variables (Table 19.1). While types ① and ② directly link options to the underlying as an external explanatory variable, the approach of type ③ first analyzes the *internal* relationships and might then also additionally consider external variables. These external explanatory variables could of course include the underlying,

TABLE 19.1 Correlation of the First Three Factors of a PCA on the Vega Sector of the JPY Volatility Surface versus Some Candidates for Explaining Variables

	SPX	VIX	Oil	BSW	5Y swap rate	Vol of vol
Factor 1	-0.73	0.52	-0.70	0.06	0.39	0.07
Factor 2	0.37	-0.34	0.34	-0.16	-0.75	-0.44
Factor 3	-0.11	-0.14	-0.02	0.15	0.21	-0.36

and in the case of Japan it turns out that the yield level is in fact significantly correlated to factor 2 (confirming the impression from the statistical analysis that it is a rather macroeconomic strategy). Factor 1 shows a high degree of correlation to “risk” variables such as stock and commodity prices. Together, this means that demand for risk assets determines the overall level of volatility, affecting both dimensions of the volatility surface (factor 1). The overall level of underlying swap rates affects the differentiation between expiries (i.e. the dimension of factor 2). The differentiation between underlying swap maturities (i.e. the dimension of factor 3), on the other hand, seems to be largely uncorrelated to external macroeconomic variables, which is in line with its statistical properties as a rather ‘pure’ relative value factor.

Again, the picture is different for other currencies. For example, at times factor 2 can be linked to the volatility of volatility, a result which might be used to replace the variable “volatility of volatility”, which is of importance in many pricing models (SABR in particular) but does not trade, with the tradable and hedgeable variable “factor 2.”

Given the statistical results and their economic interpretation, we would see no compelling relative value trade in the vega sector of the JPY volatility surface at the moment. The relative value factor 3 is close to its mean, and factors 1 and 2 represent macroeconomic events. In order to illustrate the way to construct a PCA trade on the vega part of the volatility surface, however, let’s assume that we believe that factor 2 is too high (Figure 19.25) and will decrease in the future, perhaps because we expect interest rates to increase. Let’s also assume that we have some good reason to expect the correlation with factor 1 will no longer continue.

In this case, we can look at the 1-factor residuals shown in Figure 19.26 and conclude that a short 2Y5Y versus long 10Y5Y straddle offers the best (though still rather small) profit potential. Hedge ratios are calculated according to the general PCA concept. The result is a position on the implied volatility curve (in the dimension of different expiries), hedged against changes to the overall level of implied volatility and its impact on the volatility surface (as given by the first eigenvector). If a hedge against factor 3 is desired as well (e.g. in case of the third eigenvalue being relatively high), expressing the position on factor 2 requires three instruments.

In conceptual terms, option trades of type ③ are hedged against the first factor (i.e. against changes to the overall level of implied volatility and its impact on the volatility surface). In contrast, types ① and ② hedge options against the underlying (delta hedging instead of vega hedging). Thus, the general approach of type ③ to treat implied volatility as an abstract statistical time series is correctly reflected in the choice of both the analytical tool and the hedge ratios of the trade execution. Note again that the low exposure of straddles in the vega sector to changes in the underlying is the basis that allows us

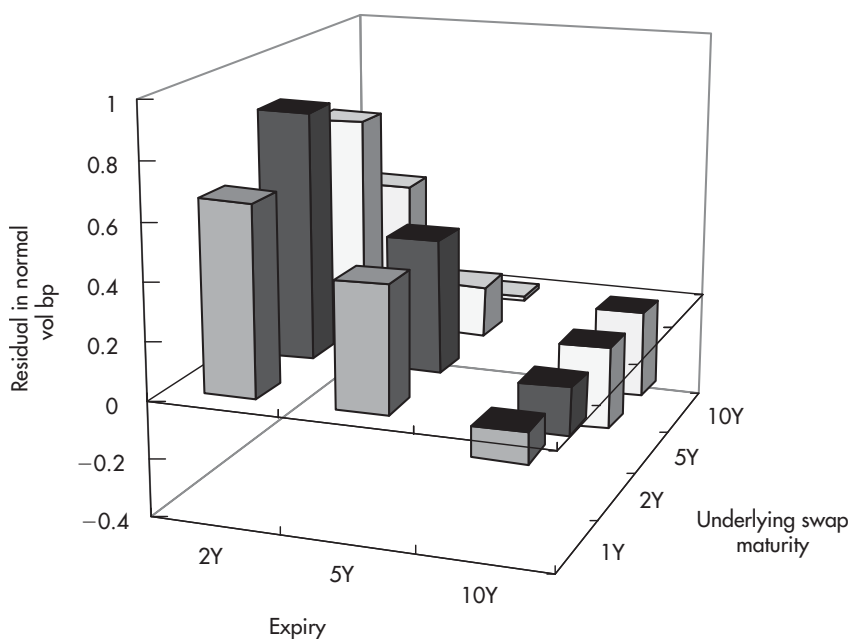


FIGURE 19.26 1-factor residuals of a PCA on the vega sector of the JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 19 Sep 2011.

to treat that part of the volatility surface conceptually by PCA statistics and to hedge against the implied volatility level rather than delta hedging.

Given the link of factor 2 to macroeconomic variables, a relative value analyst could be tempted to trade factor 2 *versus* the swap rate, thereby solving a number of problems. He obtains a relative value position hedged against the level of the macro variable “swap rate” (and against the level of the macro variable “factor 2”), which could well have a much better speed of mean reversion than factor 2. The basis for this trade is shown in Figure 19.27.

Unfortunately, the current point is close to the regression line and the relative value trade between factor 2 and the 5Y swap rate offers little profit potential (just like the “relative value factor” 3 in Figure 19.25). Moreover, there seems to be a break in the relationship, with the points in Figure 19.27 lying on two separate regression lines. All in all, we have run the right tools and tried hard, but the vega sector of the JPY option market simply does not provide a good relative value opportunity at the moment. As we have used

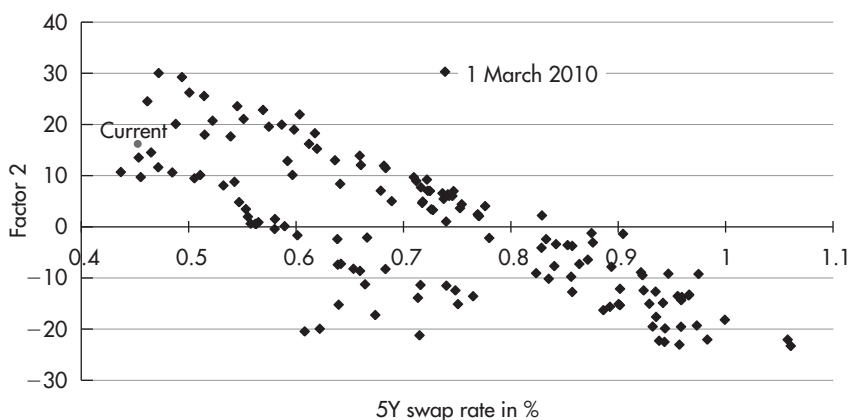


FIGURE 19.27 Factor 2 as a function of the 5Y swap rate.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data; “current” market data as of 19 Sep 2011.

throughout the book mainly examples that illustrate profitable trading strategies revealed through relative value analysis, we found it fair to finish with one example in which a proper relative value analysis leads to the correct conclusion that there are no good relative value trades. This is the more frequent experience relative value analysts will encounter, but which will in turn keep them on the lookout for the few really exceptional opportunities.

Imagine an analyst who had set up the models developed here and run them on a daily basis. After many disappointing days, on 1 March 2010 a glance at Figure 19.27 would have given him an excellent relative value trade, returning a high profit within less than a week.

PITFALLS OF OPTION TRADES OF TYPE ③

Treating options in the vega sector through a statistical model like PCA can be justified conceptually, as above, but involves a number of potential problems:

- While a one-dimensional yield curve usually has a clear factor structure, with the second eigenvalue being much larger than the third (Figure 3.4), the two-dimensional volatility surface leads to the second and third eigenvalue being often of similar magnitude (unlike in Figure 19.22, which depicts the specific situation for Japan). The reason is that both the second and third factors of a PCA on the volatility surface represent

curve steepness, just in two different dimensions. By contrast, a PCA on the one-dimensional yield *curve* contains all steepness information in the second factor only, with the third factor representing the (much less influential) curvature information in the same dimension. Therefore, higher-order eigenvectors tend to be less stable on the volatility surface than on yield curves. Moreover, the second and third factors can change their relative importance, which is a problem for PCA-hedged volatility curve trades.

- The ability to hedge against the first factor (i.e. the implied volatility level rather than delta hedging against the underlying) is based on the precondition that no delta hedging is required due to the low exposure of straddles in the vega sector to changes in the underlying. Over long periods of time, however, large moves in the underlying could violate that precondition and introduce an additional, unhedged P&L component.
- While theta is low in the vega sector, it could start to matter when holding horizons become very long. This is particularly the case when the thetas of the options involved are significantly different (e.g. in trades on a volatility curve with the same underlying and different expiries).

In general, it is important for any trade based on statistical tools to compare the time horizon over which mean reversion is expected to occur with the stability of statistical relationships. Just as in the case of statistical trades on the CDS curve, it is acceptable to ignore potential problems if the speed of mean reversion is significantly faster than the time horizon over which the assumed statistical relationships could break down. Applying these techniques can also mitigate the potential problems involved in the statistical analysis for option trades of type ③ (such as different thetas in the case of a factor 2 trade), since all three pitfalls are rather long-term issues that are less likely to affect trades with a short holding horizon. We therefore recommend restricting option trades of type ③ to such positions, whose speed of mean reversion suggests performance well before the problems described above are expected to become relevant.

CONCLUSION: SUMMARY OF OPTION TRADE TYPES AND THEIR DIFFERENT EXPOSURE

In general, we can summarize the exposure of the different types of option trades as follows. With only one underlying, all three types are rather fundamental positions, though exposed to a different fundamental variable:

- Type ① with a single underlying is exposed to the absolute level (direction) of the underlying.

- Type ② with a single underlying is exposed to the absolute level (direction) of the realized volatility.
- Type ③ with a single underlying is exposed to the absolute level (direction) of the implied volatility.

Adding at least a second underlying allows hedging against that fundamental exposure to the absolute level (direction) of a variable highly influenced by macroeconomic events and thereby constructing relative value positions. In particular:

- Type ② with (at least) two underlyings is hedged against the absolute level of *realized* volatility (and not exposed to changes in the level of implied volatility by nature).
- Type ③ with (at least) two underlyings is hedged against the absolute level of *implied* volatility (and essentially not exposed to the level of realized volatility).

Thus,

- Type ② with (at least) two underlyings is a relative value trade on the *realized* volatility curve.
- Type ③ with (at least) two underlyings is a relative value trade on the *implied* volatility curve.

SOME REMARKS ABOUT ASIAN OPTIONS

As a consequence of the transition from LIBOR term rates to overnight reference rates, Asian options, which used to be an exotic rarity, have become a standard feature of some of the most common markets. For example, during their reference month, the payoff of options on 1M SOFR futures is path-dependent. From one day to the next, i.e. at the start of the reference month, the same option on the same SOFR contract transmogrifies from a standard (American) option into an exotic Asian option (of American type). Since financial instruments are priced on averages of overnight reference rates, these “traps” have become a ubiquitous feature of many option markets – a consequence of the transition which was probably unintended by regulators and which is still ignored by some market participants.

We refer to Chapter 5 of Huggins and Schaller (2022) for a comprehensive treatment of this subject and highlight only a few key results:

- Before the reference period of an instrument settled to an average of overnight rates starts, the instrument aggregates all overnight rates into

a single variable, i.e. its price, and hence options on it can be considered and priced as standard options, similar to options on a forward term rate.

- By contrast, during the reference period, the aggregation has ended, and the options have undergone a metamorphosis into path-dependent Asian options.
- Pricing Asian options is a major challenge, which has been solved by replicating the Black–Scholes approach for those of European type settling to a geometric average under the assumption of continuous Brownian motion. While a huge academic effort is underway, we know of no complete solution for other specifications, such as the American type or settling to arithmetic averages. However, some important partial solutions are available, such as Fusai and Meucci (2008).
- The selection of the model used to price Asian options, i.e. the decision whether to include terms for jumps, drift, and diffusion, has a major impact on the results (see Table 5.2 of Huggins and Schaller, 2022). This is in line with the general statement about modeling in Chapter 6 and adds a layer of complexity already before considering the question, whether a (partial) solution exists under the model assumptions.

Relative Value in a Broader Perspective

INTRODUCTION

After focusing on the technical aspects of relative value analysis and trading, we conclude our discussion by taking a broader perspective on the macroeconomic functions of relative value analysis in society. At a time when professionals in the financial services industry increasingly need to justify their role in society, we believe it's useful to offer a few thoughts about the role of arbitrage in society.

THE MACROECONOMIC ROLE OF RELATIVE VALUE ANALYSIS AND TRADING

The term *standard of living* means different things to different people. A macroeconomist might think of GDP per person when he hears this term. Parents might think of the infant mortality rate, while retirees might think of expected longevity. Teachers might think of literacy rates, and doctors might think of rates of disease, such as tuberculosis, cholera, and malaria.

But whatever one thinks about when one hears this term, one thing stands out when comparing standards of living across societies and over time: societies that rely on free markets to organize economic activity tend to have higher standards of living than societies that rely on central planning to allocate scarce resources.

Skeptics might raise the examples of Denmark or Sweden, with their generous social models and expansive public sectors. Or they may cite the example of China, with its rapid growth rates under state capitalism. But China didn't see living standards increase until Deng made it acceptable to harness the market economy in China, and the examples of East and West Germany and of North and South Korea are usually sufficient to persuade all but the most ideological skeptics.

Market economies tend to deliver higher standards of living because prices determined in free markets act as reliable signals with which market participants can identify the best opportunities for scarce capital to improve productivity, which in the end is responsible for increases in standards of living.

For prices to act as reliable signals, they need to be efficient, in the sense that they need to reflect information that is available and relevant, particularly information pertaining to the supply and demand of the item in question. Much has been written about whether financial markets are informationally efficient – but there's little disagreement that financial markets should be informationally efficient if we want an effective allocation of capital. If a financial market was informationally inefficient, we would expect capital to be allocated in a way that resulted in few improvements in productivity.

The need for market efficiency brings us to the role of the arbitrageur, who makes a living by identifying and exploiting misvaluations among tradable instruments. As we noted in the introduction, a market characterized by too little arbitrage capital and too few arbitrageurs is a market that is likely to be plagued by market inefficiency, mispriced instruments, and ultimately an allocation of capital that does little to improve living standards in society.

While it might be an exaggeration to claim that arbitrageurs are responsible for improvements in living standards directly, arbitrageurs do play an important role in improving market efficiency and in reducing the incidence of mispriced securities – functions that are critical if capital in society is to be allocated efficiently.

Critics argue that many arbitrageurs are uninterested in living standards in society and act only in their own self-interest. To this, we have two retorts. First, the prevalence of charitable giving among hedge fund managers calls into question whether this characterization is accurate.

But second is the observation of Adam Smith: "It is not from the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest".¹ Idealists may wish the world were other than Smith observed it in 1776, but until they find a better way to enlighten the masses than did Mao and Lenin, we should consider ourselves fortunate to have an economic system that marshals self-interest toward pursuits that provide others with meat, bread, beer – and greater efficiency in capital markets.

In a capitalist society, the economic benefit of an individual is linked to the economic benefit that individual's actions provide for society. The profit of arbitrageurs can therefore be considered as the fee for their service of supporting optimal capital allocation across the productive purposes of a society.

¹From Chapter 2 of *An Inquiry into the Nature and Causes of the Wealth of Nations*, by Adam Smith, published in 1776.

ARBITRAGEURS AND POLITICIANS

We acknowledge that there is a role for government intervention to ensure the functioning of free markets *in general* (e.g. preventing market failures or monopolies on natural resources) and limit our discussion to excessive interventions pursuing *particular* political objectives.

One result of excessive government intervention is very likely to be a market mispricing, potentially even an arbitrage opportunity. And the economic cost of this mispricing is likely to be a less optimal allocation of capital, with implications for productivity, growth, and living standards.

The market mispricing caused by excessive government intervention, like all mispricings in the market, is likely to be identified and exploited by arbitrageurs. And if there are enough arbitrageurs with sufficient arbitrage capital, the mispricing is likely to be mitigated and perhaps even eliminated, resulting in an undistorted set of prices that leads to improved capital allocation and ultimately to higher living standards.

The good news in this case is that society avoids the permanently higher costs associated with inefficient capital markets. But the bad news is that the government has created a mispricing in the market that has led to even greater profits for arbitrageurs. Of course, the source of the profit in this case is the government. And in most societies, this means that the source is taxpayers. The result is that the government has transferred capital from society in general to arbitrageurs in particular. In fact, the mispricing as expressed by relative value relationships (i.e. the difference between the politically desired market equilibrium and the natural market equilibrium) is a measure of the ultimate cost of the interventionist strategy in the presence of arbitrage.

Arbitrage transforms the economic cost of excessive intervention from an inefficient capital allocation permanently affecting the whole society to a one-off cost to taxpayers from being on the losing side of arbitrage trades. Thus, in the presence of state intervention, the benefit of arbitrage to society is that it maintains functioning capital markets at a relatively low cost.

Let's consider some of the repercussions of excessive government intervention in greater detail:

- (1) The mispricing is backed by taxpayers and central banks and can thus become larger and last longer than usual.
- (2) The beneficial function of arbitrage for society appears now to be in opposition to political goals.
- (3) Politicians have an incentive to blame their losses on arbitrageurs, concealing their beneficial function for society.

With regards to (1), it may be helpful to recall Mises' argument, that partial political impacts on the market (like keeping yields artificially below their